

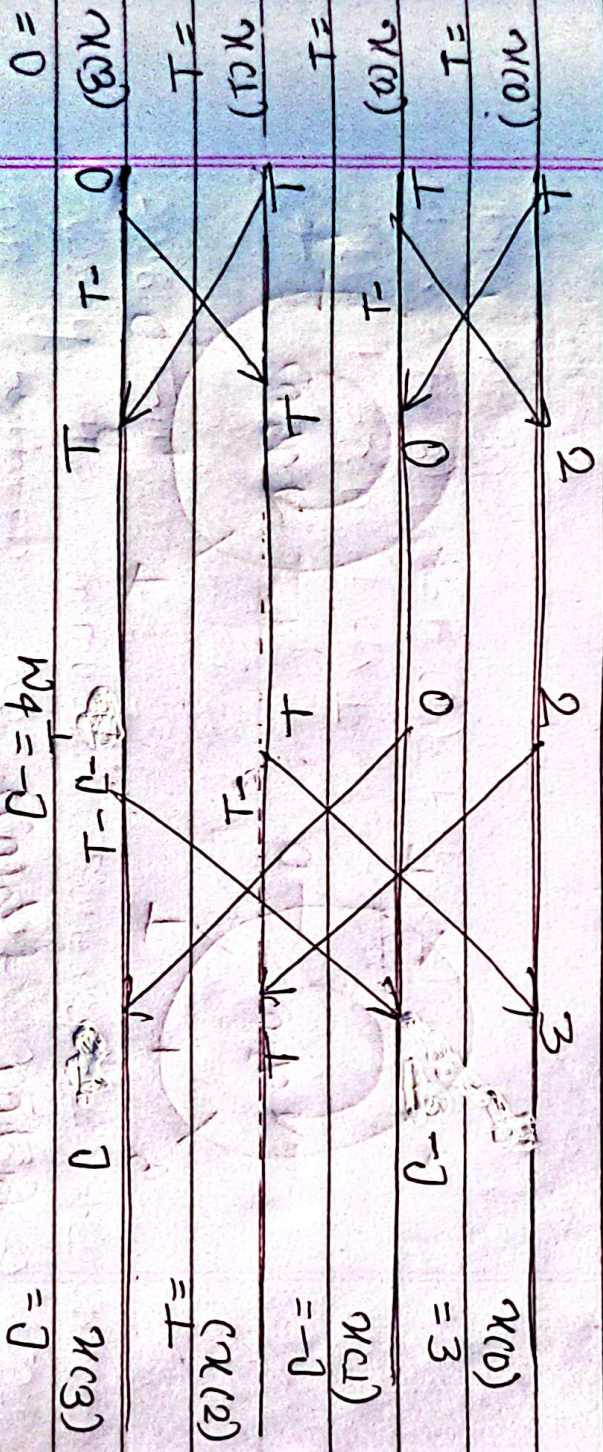
Topic 4:

Chapter 7: Discrete Fourier Transform

1. Determine the 4-Point DFT of the signal $x[n] = u[n] - u[n-3]$

$$x[n] = \begin{cases} 1 & 0 \leq n \leq 2, \\ 0 & \text{otherwise} \end{cases}$$

$$x[n] = [1, 1, 1, 0]$$



2. Compute the circular convolution of the following sequences.

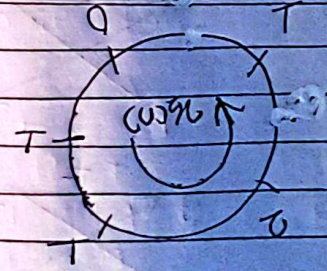
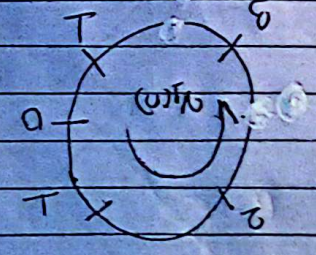
$$x[n] = [2, 0, 1, 2, 3, 1, 3]$$

$$h[n] = [5, 1, 1, 2, 1, 3]$$

1	0	1	2	3	1	1
2	0	1	2	3	1	1
3	0	1	2	3	1	1
4	0	1	2	3	1	1
5	0	1	2	3	1	1
6	0	1	2	3	1	1
7	0	1	2	3	1	1

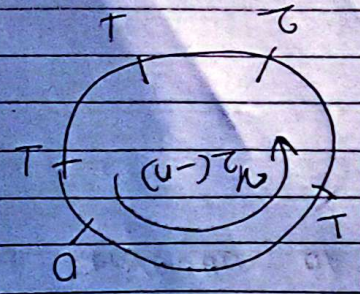
then
 $\chi_1(n) = 5, 0, 1, 2, 3, 1, 3$
 $\chi_2(n) = 5, 1, 1, 2, 1, 3$
 $\chi_3 = 5 + 4 - 1$
 $= 5 - 1$
 $= 8$

$\chi_1(m) = 5, 0, 1, 2, 3, 1, 3$
 $\chi_2(n) = 5, 1, 1, 2, 1, 3$
 $\chi_3(n) = 5, 1, 1, 2, 1, 3$

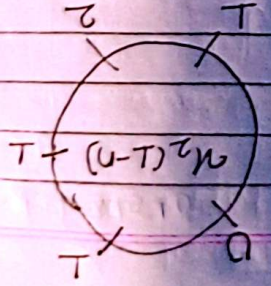


grouping $(\chi_2(n))$

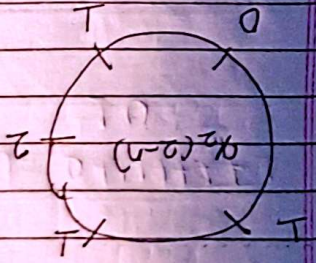
$\sum_{T=N}^{a=N} \chi_2(n) \chi_1(n)$
 $T+9+2+0+0 = [0]_N$
 $6 =$



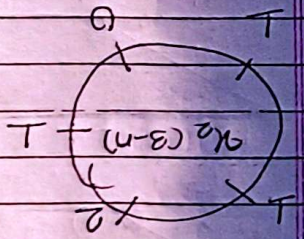
$\chi_1(n) = 0 + 1 + 0 + 3 + 2$
 $= 6$



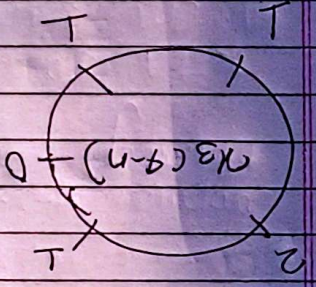
$\chi_1(n) = 0 + 1 + 2 + 0 + 1$
 $= 5$



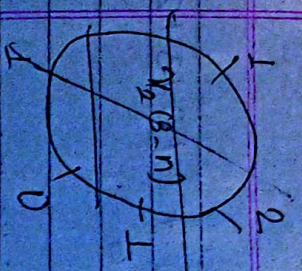
$\chi_1(n) = 0 + 2 + 2 + 3 + 0$
 $= 7$



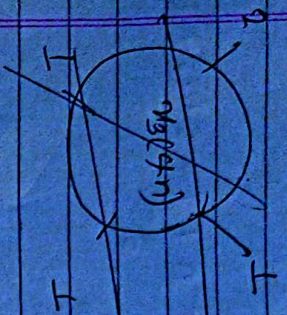
$\chi_1(n) = 0 + 1 + 4 + 3 + 1$
 $= 9$



Here, it could be convolution is $(9, 6, 5, 2, 9)$



$$x(8) = 0.12 + 3.15 + 10 = 10$$



$$x(9) = 0.12 + 9 + 10 = 10$$



Q3

Use the butterfly program to compute 8-point DIT-FFT of the following sequence using radix-2 decimation in time algorithm.

$$x(n) = \{1/2, 1/2, 1/2, 1/2, 0, 0, 0, 0\}$$

$$W_N^k = e^{-j \frac{2\pi k n}{N}}$$

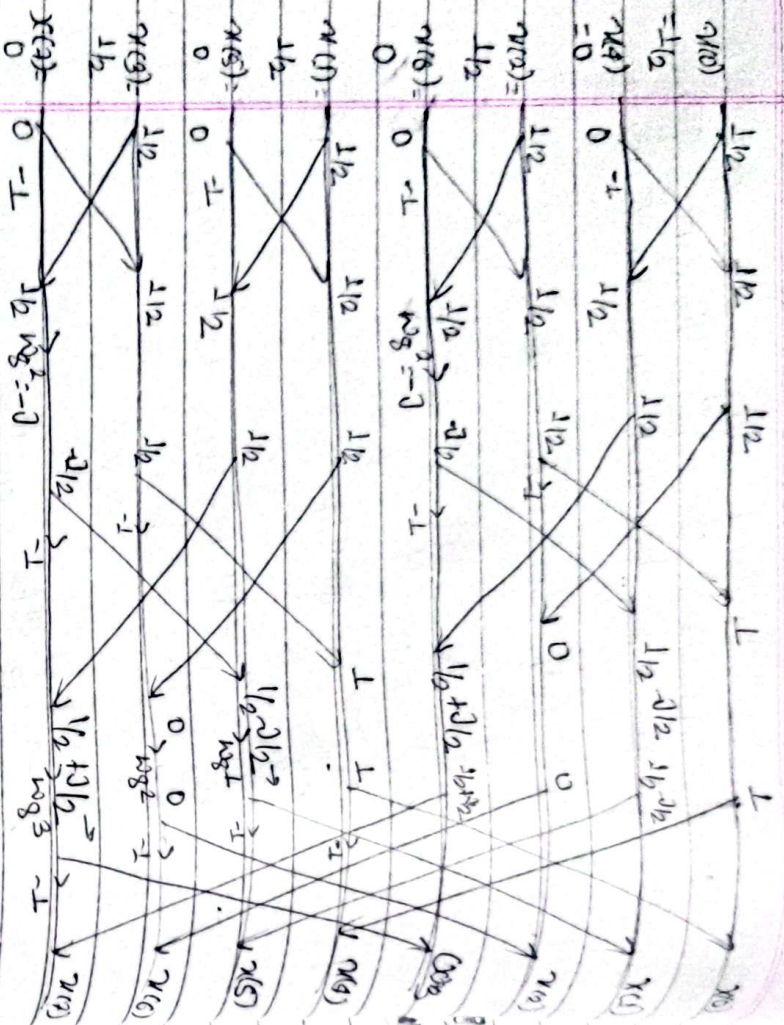
$$W_8^k = e^{-j \frac{2\pi k n}{8}}$$

$$W_8^1 = e^{-j \frac{2\pi n}{8}} = \cos \frac{\pi n}{4} - j \sin \frac{\pi n}{4} = \frac{1}{\sqrt{2}} - j \frac{1}{\sqrt{2}}$$

$$W_8^2 = e^{-j \frac{4\pi n}{8}} = \cos \frac{\pi n}{2} - j \sin \frac{\pi n}{2}$$

$$W_8^3 = e^{-j \frac{6\pi n}{8}} = \cos \frac{3\pi n}{4} - j \sin \frac{3\pi n}{4}$$

$$= \frac{1}{\sqrt{2}} - j \frac{1}{\sqrt{2}}$$



$$X(0) = 2$$

$$X(1) = \frac{1}{2} - j \left(\frac{1}{2} + \frac{1}{\sqrt{2}} \right)$$

$$X(2) = 0$$

$$X(3) = \frac{1}{2} + j \left(\frac{1}{2} - \frac{1}{\sqrt{2}} \right)$$

$$X(4) = 0$$

$$X(5) = \frac{1}{2} - j \left(\frac{1}{2} - \frac{1}{\sqrt{2}} \right)$$

$$X(6) = 0$$

$$X(7) = \frac{1}{2} + j \left(\frac{1}{2} + \frac{1}{\sqrt{2}} \right)$$

$$\log_2 \frac{1}{\sqrt{2}} = \frac{1}{2} - \frac{j}{2}$$

$$\left(\frac{1}{\sqrt{2}} \quad -\frac{j}{\sqrt{2}} \right) X \left(\frac{1}{2} \quad -\frac{j}{2} \right)$$

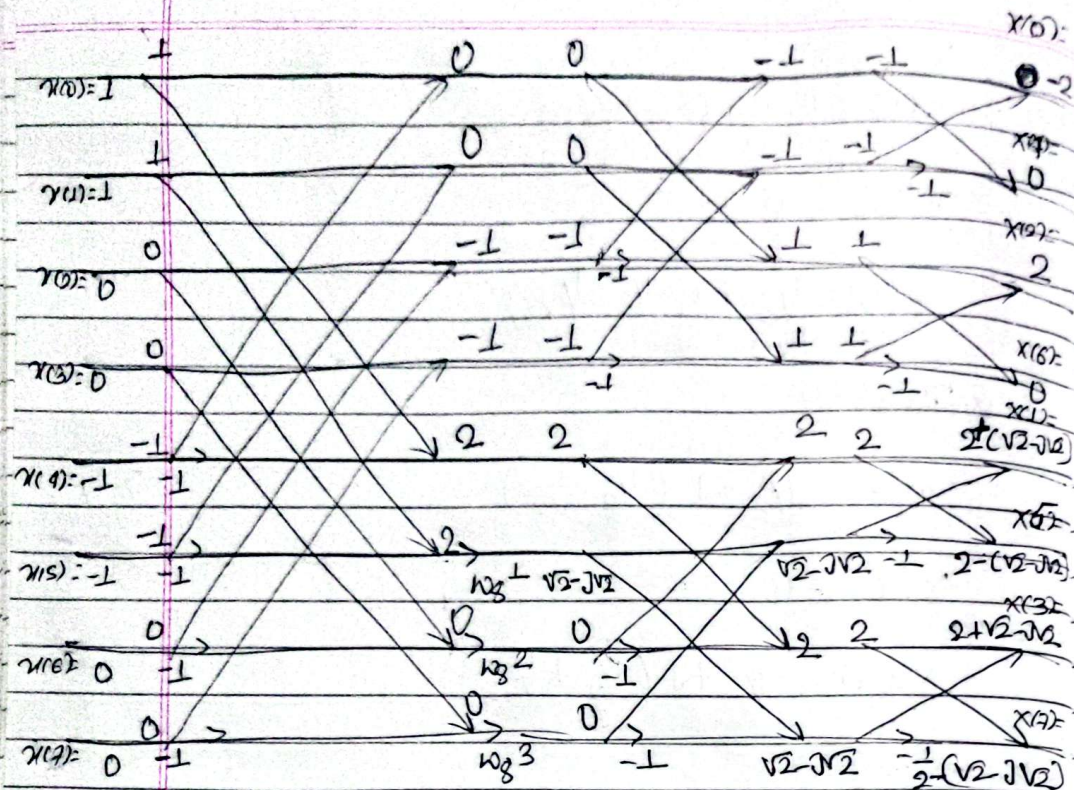
Bough

Use the butterfly diagram to compute 8-Point FFT of the following sequence using radix-2 decimation in frequency algorithm.

$$x[n] = \{1, 1, 0, 0, -1, -1, 0, 0\}$$

$$= \frac{1}{2\sqrt{2}} - \frac{j}{2\sqrt{2}} - \frac{j}{2\sqrt{2}} + \frac{j^2}{2\sqrt{2}}$$

$$= \frac{1}{2\sqrt{2}} - \frac{2j}{2\sqrt{2}} + \frac{j^2}{2\sqrt{2}}$$



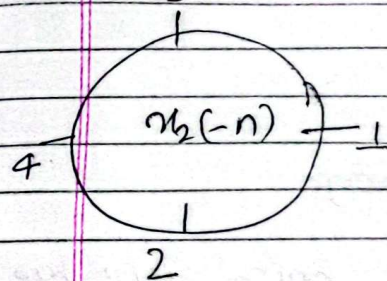
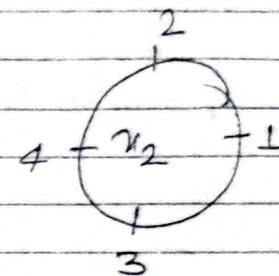
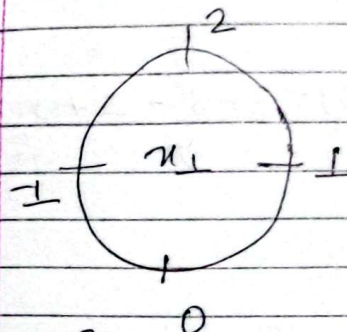
$$w_8^1 = \frac{1}{\sqrt{2}} - j \frac{1}{\sqrt{2}}$$

$$w_8^2 = -j$$

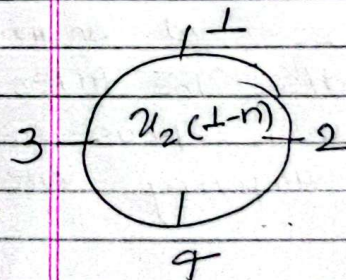
$$w_8^3 = -1/\sqrt{2} - j1/\sqrt{2}$$

Q5 A FIR Filter with Impulse response $h[n] = \{1, 2, 4, 3\}$ is excited with the input sequence $x[n] = \{1, 2, -1\}$. Use circular convolution using ASCII graphs to compute the output of the filter.

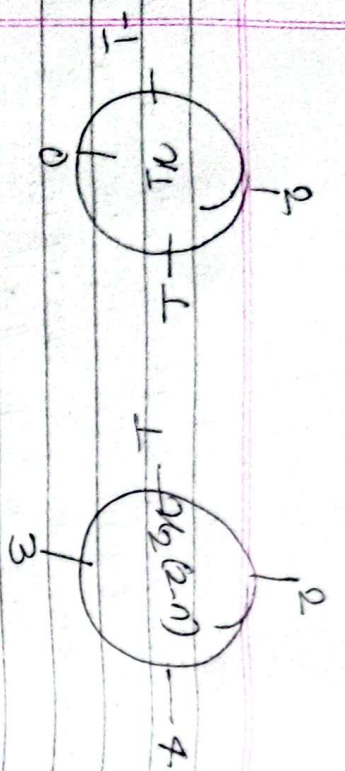
$x_1[n] = \{1, 2, -1\} \rightarrow$ zero padding
 $x_2[n] = \{1, 2, 4, 3\}$ $\{1, 2, -1, 0\}$



$$y[0] = 1 + 6 + (-4) + 0 = 3$$



$$y[1] = 2 + 2 + (-3) + 0 = 1$$



$$x[2] = 4 + 4 - 1 + 0 = 7$$

$$x[3] = 3 + 8 - 2 + 0 = 9$$

Circular convolution = {3, 4, 7, 9}

Tutorial 5:

Chapter 6 FIR Filter Design

- Design a lowpass filter which will have -3dB cut off at 30π rad/sec and an attenuation of 50dB at 45π rad/sec. The filter is required to have a linear phase and the system uses a sampling rate of 100 samples/sec.

Soln:—

Here,

Cutoff frequency (ω_c) = 30π rad/sec

Stopband frequency (ω_s) = 45π rad/sec

Sampling rate (f_s) = 100 samples/sec

Stopband attenuation (A_s) = 50 dB

Now,

After sampling,

$$\omega_c = \frac{30\pi}{100} = 0.3\pi$$

$$\omega_s = \frac{45\pi}{100} = 0.45\pi$$

Hamming window

$$H_d(\omega) = e^{-j\omega T}, \quad -\omega_c \leq \omega \leq \omega_c$$

$$= 0, \text{ otherwise}$$

Now applying IDFT,

$$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{-j\omega T} e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega(n-T)} d\omega$$

when

$n \neq T$

$$= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} \frac{e^{j\omega(n-T)}}{j(n-T)} d\omega$$

$$= \frac{1}{2\pi j(n-T)} \left[e^{j\omega(n-T)} - e^{-j\omega_c(n-T)} \right]$$

$$= \frac{1}{j\pi(n-T)} \left[\sin \omega_c(n-T) \right]$$

$$= \frac{\sin \omega_c(n-T)}{\pi(n-T)}$$

when

$n = T$

$$h_d(n) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} d\omega$$

$$= \frac{1}{2\pi} [\omega]_{-\omega_c}^{\omega_c} = \frac{\omega_c}{\pi}$$

$$h_d(n) = \frac{\sin \omega_c(n-T)}{\pi(n-T)} ; n \neq T$$

$$h_d(n) = \frac{\omega_c}{\pi}, n = T$$

Transition width of main lobe of sampling window is $\frac{8\pi}{M}$

General formula of lobe width = $K \left(\frac{2\pi}{M} \right)$

Now,

$$\frac{8\pi}{M} = K \frac{2\pi}{M}$$

$$\therefore K = 4$$

Now,

order is,

$$N = K \left(\frac{2\pi}{\omega_1 - \omega_2} \right)$$

$$= 4 \left(\frac{2\pi}{0.45\pi - 0.3\pi} \right)$$

$$= 53.33$$

$$N \approx 55$$

$$\therefore T = \frac{55-1}{2} = 27$$

Now,

$$h_d(n) = \frac{\sin 0.3\pi(n-27)}{\pi(n-27)} ; n \neq 27$$

$$= 0.3, n = 27$$

W.D.W,

$$h(n) = h_d(n) \cdot w(n)$$

$$w(n) = 0.54 - 0.46 \cos \frac{2\pi n}{M-1}$$

$$= 0.54 - 0.46 \cos \frac{2\pi n}{55-1}$$

$$= 0.54 - 0.46 \cos \frac{\pi n}{27}$$

$$h(n) = h_d(n) \cdot w(n)$$

$$= \frac{\sin 0.3\pi(n-27)}{\pi(n-27)} * 0.54 - 0.46 \cos \frac{\pi n}{27}, n=27$$

$$= 0.3 \cos - 0.46 \cos \frac{\pi n}{27}, n=27$$

(3) Design an FIR filter those filter using Kaiser window to meet the following specifications

$$0.99 \leq |H(e^{j\omega})| \leq 1.01, \text{ for } 0 \leq |\omega| \leq 0.19\pi$$

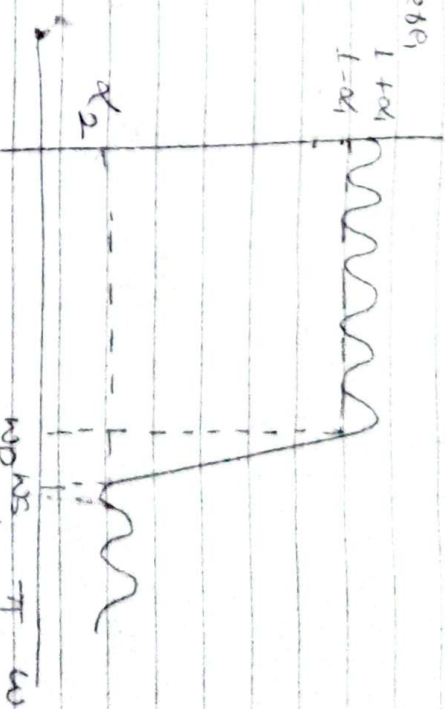
$$|H(e^{j\omega})| \leq 0.01, \text{ for } 0.21\pi \leq |\omega| \leq \pi$$

Soln-

HPB,

$$1 + \alpha_1$$

$$1 - \alpha_1$$



from figure,

$$\omega_p = 0.19\pi$$

$$\omega_s = 0.21\pi$$

$$\alpha_2 = 0.01$$

$$\alpha_1 = 0.01$$

$$1 + \alpha_1 = 1.01$$

$$1 - \alpha_1 = 0.99$$

$$\alpha_1 = 0.01 = \alpha_2$$

$$\alpha_1 = 0.01$$

Transition width $(\Delta\omega) = \omega_s - \omega_p$

$$= 0.02\pi$$

$$\omega_c = \frac{\omega_p + \omega_s}{2}$$

$$= \frac{0.19\pi + 0.21\pi}{2}$$

$$= \frac{0.40\pi}{2}$$

$$= 0.2\pi$$

Minimum Error

$$\alpha = 0.01$$

$$A = -20 \log_{10} \alpha$$

$$= -20 \log_{10} 0.01$$

$$= 40 \text{ dB}$$

$$B = 0.5392 (40 - 21) + 0.02886 (40 - 21)$$

$$= 3.3953$$

$$M = \frac{A - B}{2.285 \text{ dB}}$$

$$= 222.88$$

$$M \approx 224 \text{ (more even)}$$

Now,

$$\alpha = \frac{M}{2}$$

$$= \frac{224}{2} = 112$$

For Kaiser window,

$$w(n) = \begin{cases} I_0 \{ 3.395 [1 - (\frac{n-112}{112})^2]^{1/2} \} & 0 \leq n \leq 112 \\ 0 & \text{otherwise} \end{cases}$$

I_0 : Bessel's Function

$$I_0 x = 1 - \frac{0.25}{x} + \dots$$

Now,

$$H_d(\omega) = \begin{cases} e^{-j\omega T} & -\omega_c \leq \omega \leq \omega_c \\ 0 & \text{otherwise} \end{cases}$$

on solving $H_d(\omega)$, we obtain i.e. taking inverse fourier transform of $H_d(\omega)$

$$\therefore h_d(n) = \begin{cases} \frac{\sin \omega_c (n-T)}{\pi (n-T)} & \text{for } n \neq T \\ \omega_c & \text{for } n = T \end{cases}$$

$$\text{Since, } T = \frac{M}{2} = 112$$

$$\omega_c = 0.2 \pi \text{ rad/sec}$$

$$\therefore h_d(n) = \begin{cases} \frac{\sin 0.2 \pi (n-112)}{\pi (n-112)} & \text{for } n \neq 112 \\ 0.2 & \text{for } n = 112 \end{cases}$$

Now,

$$h(n) = h_d(n) \cdot w(n)$$

$$= \begin{cases} \frac{\sin 0.2 \pi (n-112)}{\pi (n-112)} \times I_0 \{ 3.3953 [1 - (\frac{n-112}{112})^2]^{1/2} \} & n \neq 112 \\ 0.2 \times 0 & n = 112 \end{cases}$$

2. A low pass filter is described to have frequency response $H_d(\omega)$ as defined below. Design FIR filter using Blackman window to meet the following requirement, 1- the cut-off frequency $\omega_c = \pi/3$ and order $N=5$. Does this filter have linear phase?

$$H_d(\omega) = \begin{cases} e^{-j\frac{(N-1)\omega}{2}} & 0 \leq |\omega| \leq \omega_c \\ 0 & \text{otherwise} \end{cases}$$

The Blackman window is defined as,

$$w(n) = \begin{cases} 0.42 - 0.5 \cos \frac{2\pi n}{N-1} + 0.08 \cos \frac{4\pi n}{N-1} & 0 \leq n \leq N-1 \end{cases}$$

Soln:— $M=N$

$M=5$

$\omega_c = \pi/3$

$L = \frac{M-1}{2} = \frac{5-1}{2} = 2$

$\therefore T = 2$

Now, using IDTFT

$$h_d(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(\omega) e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{-j\frac{(N-1)\omega}{2}} \cdot e^{j\omega n} d\omega$$

$$= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{\frac{-j(N-1)\omega}{2} + j\omega n} d\omega$$

$$= \frac{2 \sin \left(\frac{n - \frac{(N-1)}{2}}{2} \right) \omega_c}{2\pi} \cdot \omega_c$$

$$\frac{\sin \omega_c \left[\frac{n - \frac{(N-1)}{2}}{2} \right]}{\pi}$$

$$\frac{\sin \left(\frac{n - \frac{(N-1)}{2}}{2} \right)}{\pi}$$

Where $n = T$
Then,

$$h_d(n) = \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} \frac{\sin \left(\frac{n - \frac{(N-1)}{2}}{2} \right)}{\pi} d\omega = 0$$

$h_d(n) = \frac{\omega_c}{\pi} \cdot n \leq T$

$\frac{\omega_c}{\pi} \sin \left(\frac{n - \frac{(N-1)}{2}}{2} \right) \omega_c$
 $n \neq T$

0 1 2 3 4

1 2 3 4

1 1 1 1

$$h(n) = \frac{\sin \omega_c}{\pi} \left[\frac{n - (N-1)/2}{N-1} \right] ; n \neq N$$

$$= \frac{\omega_c}{6} ; n \neq N$$

$$h(n) = \sin \omega_c \left[\frac{n-1}{N-1} \right] ; n \neq 2$$

$$\frac{\omega_c}{6} ; n = 2$$

$$h(n) = h(n) \cdot \omega_c(n)$$

$$h(n) = \sin \omega_c(n-1) \cdot \frac{\omega_c(n-1)}{\pi(n-1)}$$

$$\therefore h(n) = \sin \omega_c (n-1) ; n \neq 2$$

$$= \frac{\omega_c}{\pi} ; n = 2$$

n	0	1	2	3	4
h(n)	0.138	0.296	1/3	0.296	0.138
h(n)	-0.246	0.38	0.91	0.45	-0.078

4. Explain the followings:-

1. FIR filter using rectangular window

A finite impulse response (FIR) filter is a type of digital filter that has a finite duration impulse response.

The Rectangular window method is one of the simplest techniques used to design FIR filters.

$$h(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(e^{j\omega}) e^{j\omega n} d\omega$$

For example, a low-pass filter,

$$h_d(n) = \frac{\sin(\omega_c n)}{\pi n}, n \neq 0$$

$$h_d(0) = \omega_c$$

$\therefore$ ω_c is the cutoff frequency.

Rectangular window is defined as:

$$w(n) = 1, 0 \leq n \leq N-1$$

$$w(n) = 0, \text{ otherwise}$$

The Final Fx Filter coefficients are obtained as:-

$$h(n) = h_d(n) \cdot w(n)$$

b. Gibbs Phenomenon

It is a fundamental effect in Fourier series and transform approximations of discontinuous function.

In the Fourier transform, Gibbs phenomenon appears due to the truncation of the sinc function in frequency response.

$$H_N(\omega) = \int_{-\infty}^{\infty} h_N(t) e^{-j\omega t} dt$$

$\Rightarrow$ Reducing Gibbs Phenomenon

2. Smoother windowing functions

Instead of a rectangular window, use

$$\text{Hamming window: } w(n) = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right)$$

Blackman window:-

$$w(n) = 0.42 - 0.5 \cos\left(\frac{2\pi n}{N-1}\right) + 0.08 \cos\left(\frac{4\pi n}{N-1}\right)$$

a. Linear Phase Filters

A linear phase filter is a type of filter where the phase response is a linear function of frequency.

For an FIR filter with impulse response $h(n)$, the phase response is given by:-
 $\theta(\omega) = -\angle H(\omega)$

It must satisfy symmetry or antisymmetry

1. Symmetric Impulse Response

$$h(n) = h(N-1-n), \text{ (Even symmetry)}$$

2. Anti symmetric Impulse Response

$$h(n) = -h(N-1-n), \text{ (odd symmetry)}$$

Applications

Audio Processing
Communication

Image processing

Type	symmetry	length N	Applications
I	symmetric	odd	Low-pass, high-pass, band-pass
II	"	even	Low-pass
III	Antisymmetric	odd	Hilbert transform
IV	"	even	Hilbert transform

Chapter 7: IIR Filter design

1. Derive a transfer relation of IIR filter between s-plane and z-plane by using Impulse Invariance and Bilinear transformation method.

→ By Impulse Invariance method

$$H_a(s) = \sum_k \frac{TR_k}{s - s_k}$$

$H_a(s)$:- analog transfer function in the Laplace domain

where:

s_k are the poles of $H_a(s)$

TR_k are the residues at these poles

$$h_a(t) = \sum_k TR_k e^{s_k t} u(t)$$

↓

Inverse Laplace transform of $H_a(s)$

$t = nT$ CT is the sampling period
 $t = nT$

$$h(nT) = h_a(nT) = \sum_k TR_k e^{s_k nT}$$

The Z-transform of this discrete impulse response,

$$H(z) = \sum_k R_k \frac{1}{1 - e^{s_k T} z^{-1}}$$

Thus,

the mapping from the s-plane to the z-plane is:

$$z = e^{sT}$$

"

By Bilinear Transformation method.

$$s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}$$

which maps the entire s-plane into the unit circle in the z-plane.

analog transfer function is given by:

$$H_a(s) = \frac{N(s)}{D(s)}$$

By substituting the bilinear transformation formula:

$$s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}$$

discrete transfer function:-

$$H(z) = H_a \left(\frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}} \right)$$

Feature	Impulse Invariance	Bilinear Transformation
- Aliasing	yes	No
- Frequency Mapping	$z = e^{sT}$	$s = \frac{2}{T} \frac{1 - z^{-1}}{1 + z^{-1}}$
- Frequency Distortion	No	Yes
- used for preserving original shape	Low Pass Filters	All Filter types

convert the analog filter into a digital filter whose system function is

use impulse invariant technique Assume $T =$

(a) $H(s) = \frac{s + 0.2}{(s + 0.2)^2 + 9}$ 0.1-sec

$H(s)$ is of form $\frac{s + a}{(s + a)^2 + b^2}$

$$\frac{s+a}{(s+a)^2+b^2} \rightarrow \frac{1-e^{-aT}(\cos bT)z^{-1}}{1-2e^{-aT}(\cos bT)z^{-1}+e^{-2aT}z^{-2}}$$

$$a=0.2 \quad \text{and} \quad b=3$$

$$H(s) = \frac{s+0.2}{(s+0.2)^2+9}$$

$$= \frac{s+0.2}{(s+0.2)^2+3^2}$$

$$H(z) = \frac{1-e^{-0.2T}(\cos 3T)z^{-1}}{1-2e^{-0.2T}(\cos 3T)z^{-1}+e^{-2 \times 0.2T}z^{-2}}$$

$$= \frac{1-e^{-0.2T}(\cos 3T)z^{-1}}{1-2e^{-0.2T}(\cos 3T)z^{-1}+e^{-0.4T}z^{-2}}$$

$$\text{Assume } T = 0.1 \text{ sec,}$$

$$= \frac{1-e^{-0.2 \times 0.1}(\cos 3 \times 0.1)z^{-1}}{1-2e^{-0.2 \times 0.1}(\cos 3 \times 0.1)z^{-1}+e^{-0.4 \times 0.1}z^{-2}}$$

$$= \frac{1-0.9364z^{-1}}{1-1.9228z^{-1}+0.67032z^{-2}} \quad \#$$

$$b. \quad H(s) = \frac{1}{(s+0.5)(s^2+0.5s+2)}$$

using Partial fraction,

$$\frac{A}{s+0.5} + \frac{Bs+C}{s^2+0.5s+2} \quad \text{--- (1)}$$

$$\text{Multiply } A(s^2+0.5s+2) + (Bs+C)(s+0.5)$$

$$1 = A(s^2+0.5s+2) + (Bs+C)(s+0.5)$$

$$1 = As^2+0.5As+2A + Bs^2+0.5Bs+Cs+0.5C$$

$$1 = As^2+Bs^2+0.5As+0.5Bs+Cs+2A+0.5C$$

$$1 = (A+B)s^2 + (0.5A+0.5B+C)s + 2A+0.5C$$

Now,

$$A+B=0 \quad ; \quad 0.5A+0.5B+C=0$$

$$\therefore A=B \quad -0.5+0.5B+C=0$$

$$\therefore C=0$$

$$2A+0.5C=1$$

$$-2B+0.5C=1$$

$$-2B+0=1$$

$$B = -1/2 \quad \therefore A = 1/2$$

Putting values of AB and C in eqn (1)

$$\frac{0.5}{s+0.5} - \frac{0.5s}{s^2+0.5s+2}$$

$$= \frac{0.5}{1-e^{-0.5T}} - \frac{0.5s}{s^2+0.5s+2} \quad \text{--- (A)}$$

Now, Doing for

$$= \frac{0.5s}{s^2+0.5s+2} \quad \frac{s+1}{(s+1)^2+1} \rightarrow$$

$$= \frac{0.5s}{s^2+2.5s+1.4} + \frac{(1.4)^2}{(1.4)^2+1} \cdot 12$$

$$= \frac{0.5s}{(s+1.4)^2+1.93}$$

$$= \frac{0.5s}{(s+0.25)^2+(1.39)^2}$$

$$= \frac{0.5s}{(s+0.25)^2+(1.39)^2} = \frac{0.5 \times 0.25}{1.39} \cdot \frac{1.39}{(s+0.25)^2+(1.39)^2}$$

$$= \frac{0.5}{(s+0.25)^2+(1.39)^2}$$

$$= \frac{0.5}{(s+0.25)^2+(1.39)^2} - \frac{0.5 \times 0.25}{(s+0.25)^2+(1.39)^2}$$

$$= \frac{0.5 \cdot 0.5}{1-2e^{-0.25T}} \cos(1.39T) \cdot 2^{-1} -$$

$$\frac{0.5 \times 0.25}{1.39} \times \frac{1.39}{(s+0.25)^2+(1.39)^2}$$

$$= \frac{0.5}{1-2e^{-0.25T}} \cos(1.39T) \cdot 2^{-1} - \frac{1-2e^{-0.25T}}{\cos(1.39T) \cdot 2^{-1} + e^{-0.5T} \cdot 2^{-2}}$$

$$\frac{0.08 \cdot e^{-0.25T} \sin(1.39T) \cdot 2^{-1}}{1-2e^{-0.25T} \cos(1.39T) \cdot 2^{-1} + e^{-0.5T} \cdot 2^{-2}}$$

from eqn (A)

$$\frac{0.5}{1-e^{-0.5T}} - \frac{0.5}{1-2e^{-0.25T} \cos(1.39T) \cdot 2^{-1} + e^{-0.5T} \cdot 2^{-2}}$$

$$+ \frac{0.08 \cdot e^{-0.25T} \sin(1.39T) \cdot 2^{-1}}{1-2e^{-0.25T} \cos(1.39T) \cdot 2^{-1} + e^{-0.5T} \cdot 2^{-2}}$$

let,

$$T = 0.1 \text{ sec}$$

then,

$$H(z) = \frac{0.5}{1 - 0.95z^{-1}} - \frac{0.5 - 0.96z^{-1}}{1 - 1.93z^{-1} + 0.95z^{-2}}$$

$$+ \frac{0.01z^{-1}}{1 - 1.88z^{-1} + 0.95z^{-2}} \quad \text{---} \quad H$$

3. convert the analog filter with system function

$$H(s) = \frac{s + 0.1}{(s + 0.1)^2 + 9}, \quad \omega_0 = \frac{\pi}{4}$$

Into a digital filter using bilinear transformation,

Given,

$$\omega_c \text{ (cut-off frequency)} = \pi/4$$

$$\omega_c^2 = 9$$

$$\omega_c = 3$$

$$\omega_c = \frac{2}{T} \tan \left[\frac{\omega_c}{2} \right]$$

$$3 = \frac{2}{T} \tan \left[\frac{\pi/4}{2} \right]$$

$$T = \frac{2}{3} \tan \left[\frac{\pi}{8} \right]$$

$$= 0.226 \text{ sec}$$

$$H(s) = \frac{s + 0.1}{(s + 0.1)^2 + 9}$$

using Bi-linear transformation,

$$H(z) = H(s) \quad \left| \quad s \rightarrow \frac{2}{T} \left[\frac{1-z^{-1}}{1+z^{-1}} \right] \right.$$

$$H(z) = \frac{2}{T} \left[\frac{1-z^{-1}}{1+z^{-1}} \right] + 0.1$$

$$\left[\frac{2}{T} \left[\frac{1-z^{-1}}{1+z^{-1}} \right] + 0.2 \right]^2 + 9$$

Substitute $T = 0.226 \text{ s}$

$$H(z) = \left[\left(\frac{2}{0.226} \right) \cdot \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 0.2 \right]^2 + 9$$

$$\left[\left(\frac{2}{0.226} \right) \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 0.2 \right]^2 + 9$$

$$= \left[(2.25) \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 0.2 \right]^2 + 9$$

$$\left[(2.25) \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 0.2 \right]^2 + 9$$

$$H(z) = \frac{[(7.95)(1-z^{-1}) + 0.1(1+z^{-1})]}{(1+z^{-1})}$$

$$(7.95)^2 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)^2 + 10.4z^{-1} + 2 \left[7.95 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right]$$

$$* 0.1] + 19$$

$$= \frac{[7.95^2 - 7.95z^{-1} + 0.1 + 0.1z^{-1}]}{(1+z^{-1})}$$

$$52.56 \left[\frac{(1-z^{-1})^2}{(1+z^{-1})^2} \right] + 0.01 + \left[1.95 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right]$$

$$= \frac{7.95^2 - 7.95z^{-1}}{(1+z^{-1})}$$

$$52.56 \frac{(1-z^{-1})^2}{(1+z^{-1})^2} + 9.01 \frac{(1+z^{-1})^2}{(1+z^{-1})^2} +$$

$$1.95 \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \times \frac{(1+z^{-1})}{(1+z^{-1})}$$

$$= \frac{(7.95^2 - 7.95z^{-1})}{(1+z^{-1})}$$

$$52.56 (1-z^{-1})^2 + 9.01 (1+z^{-1})^2 + 1.95 (1+z^{-1})$$

$$(1+z^{-1})^2$$

$$= \frac{(7.95^2 - 7.95z^{-1})}{(1+z^{-1})}$$

$$52.56 (1-z^{-1})^2 + 9.01 (1+z^{-1})^2 + 1.95 (1+z^{-1})$$

$$= \frac{7.95^2 - 7.95z^{-1}}{(1+z^{-1})}$$

$$52.56 (1-z^{-1})^2 + 9.01 (1+z^{-1})^2 + 1.95 (1+z^{-1})$$

$$= \frac{7.95^2 - 7.95z^{-1}}{(1+z^{-1})}$$

$$52.56 (1-z^{-1})^2 + 9.01 (1+z^{-1})^2 + 1.95 (1+z^{-1})$$

$$= \frac{7.95^2 - 7.95z^{-1}}{(1+z^{-1})}$$

$$= \frac{7.95^2 - 7.95z^{-1}}{(1+z^{-1})}$$

$$= \frac{7.95^2 - 7.95z^{-1}}{(1+z^{-1})}$$

4. Design a lowpass Butterworth filter to meet the following specifications

Passband gain = 0.89

Attenuation = 0.00

Passband frequency edge = 30 Hz
Stopband edge = 25 Hz

Soln:—

$$\omega_p = 2\pi \times 30 \cdot 60\pi \text{ rad/s}$$

$$\omega_{pass} = 30 \text{ Hz}$$

$$\omega_s = 2\pi \times 15$$

$$= 150\pi \text{ rad/s}$$

$$\omega_{stop} = 75\pi \text{ rad/s}$$

$$\alpha_{max} = -20 \log_{10} (0.89) = 1.01 \text{ dB}$$

$$\alpha_{min} = -20 \log_{10} (0.20) = 13.9794 \text{ dB}$$

$$\epsilon^2 = 10^{0.1 \alpha_{max} - 1} = 10^{0.1 \times 1.01 - 1} = 0.061827 \Rightarrow 0.5116$$

$$\delta^2 = 10^{0.1 \alpha_{min} - 1} = 10^{1.39794 - 1} = 24$$

$$\Rightarrow \delta = 4.899$$

$$\therefore N = \frac{\log_{10} \left(\frac{\delta}{\epsilon} \right)}{\log_{10} \left(\frac{4.899}{0.5116} \right)} = \frac{\log_{10} \left(\frac{150\pi}{75\pi} \right)}{\log_{10} \left(\frac{150\pi}{60\pi} \right)}$$

$$= 2.4656 \approx 3$$

$$\omega_c = \omega_s = \frac{150\pi}{4.899^{1/3}} = 277.9632 \approx 278$$

Choose lower bound
values for simplicity.

for, $N=3$ and $\omega_c=1$

$$H(s) = \frac{1}{(s+1)(s^2+s+1)}$$

$\omega_c=1$ (for normalized)

$\omega_c=278$ need to do normalized $H(s)$
by $\omega_c=278 \text{ rad/s}$

$$H(s) = \frac{s/\omega_c}{(s/\omega_c)^2 + (s/\omega_c) + 1} = \frac{\omega_c^3}{(s+\omega_c)(s^2+\omega_c s+\omega_c^2)}$$

$$= \frac{19683000}{(s+278)(s^2+278s+72900)}$$

By impulse invariance method:

$$H(s) = \frac{19683000}{(s+278)(s^2+278s+72900)}$$

$$= \frac{A}{s+278} + \frac{B+C}{s^2+278s+72900}$$

$$19683000 = As^2 + 278As + 72900A + Bs^2 + 278Bs + Cs + 278C$$

$$A+B=0$$

$$270A + 1270B + C = 0$$

$$12900A + 270C = 1968300$$

$$A = 270$$

$$B = -270$$

$$C = 0$$

$$\therefore H(s) = \frac{270}{s+270} - \frac{270s}{s^2+270s+172900}$$

$$s+270 \quad s^2+270s+172900$$

$$= \frac{270}{s+270} - \frac{270s}{s^2+270s+135^2+233.83^2}$$

$$= \frac{270}{s+270} - \frac{270(s+135)}{(s+135)^2+233.83^2} +$$

$$\frac{36450}{233.83} \times \frac{233.83}{(s+135)^2+233.83^2}$$

using mapping method:

$$H(z) = \frac{270}{1-e^{-270T}z^{-1}} - \frac{270 \times 1-e^{-135T} \cos(233.83T)}{1-2e^{-135T} \cos(233.83T)z^{-1} + e^{-270T}z^{-2}}$$

$$+ \frac{155.88 \times e^{-135T} \sin(233.83T)z^{-1}}{1-2e^{-135T} \cos(233.83T)z^{-1} + e^{-270T}z^{-2}}$$

$$e^{-270T}z^{-2}$$

by bilinear transformation method:-

$$\omega_p = \omega_{PT} = 2\pi F_{pass}/F_s$$

$$= 2\pi (30/300) = \pi/5$$

$$\omega_s = 2\pi (15/300) = \pi/2$$

Pre-warping,

$$\omega_p' = \frac{2}{T} \tan\left(\frac{\omega_p T}{2}\right) = \frac{2}{T} \times 0.3249$$

$$\omega_s' = \frac{2}{T} \tan\left(\frac{\omega_s T}{2}\right) = \frac{2}{T} \times 1$$

$$\therefore N = \frac{\log_{10}\left(\frac{\delta}{\epsilon}\right)}{\log_{10}\left(\frac{\omega_p'}{\omega_s'}\right)} = \frac{\log_{10}\left(\frac{4.899}{0.5396}\right)}{\log_{10}\left(\frac{1}{0.3249}\right)} = 1.962$$

$$\therefore N \simeq 2$$

$$\omega_c = \frac{\omega_p}{\epsilon^{1/N}} = \frac{2}{T} \times 0.3249 = 265.37 = 270$$

$$\text{for } N=2 \text{ and } \omega_c=1 \quad (\text{from table})$$

$$H(s) = \frac{1}{(s^2+1.414s+1)}$$

$$H(\omega)|_s = s/\omega_c = \frac{\omega_c^2}{(s^2 + 1.414\omega_c s + \omega_c^2)}$$

$$= \frac{72900}{(s^2 + 382.6s + 72900)}$$

mapping by bilinear transformation,

$$s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)$$

$$\therefore H(z) = \frac{72900}{\left(\frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) \right)^2 + 382.6 \times \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + 72900}$$